

NanoGT Match Report: Angelman syndrome

Disease: Angelman syndrome (ORPHA:72)

Primary gene: UBE3A

Gene CDS: 2598 bp

Inheritance: Autosomal dominant

Target tissues scored: CNS

Interpretation

- No high-confidence vector precedent was found; the best result is medium-confidence and should be treated as manual-review territory.
- Main review flags: Inheritance/mechanism may not be simple loss-of-function replacement; check dominant-negative, gain-of-function, or mitochondrial biology; Mechanism evidence is conditionally compatible with gene addition; review the listed disease-specific constraints before treating vector precedent as transferable; Dominant inheritance flagged; assess whether silencing, editing, or allele-specific strategy is needed instead of simple addition.

Disease Mechanism Evidence

Molecular mechanism: genomic imprinting

Mechanistic detail: Maternal UBE3A is lost by deletion or mutation; paternal UBE3A is epigenetically silenced in neurons by the SNHG14/UBE3A-ATS antisense transcript; neuronal UBE3A expression is absent from both alleles

Gene-addition compatibility: conditional

Preferred modality class: aav neuronal ube3a transgene addition

Evidence level/status: direct / source_checked

Evidence summary: Mabb AM et al. (2011 Trends Neurosci PMID 21592595) described the dual-allele loss mechanism — maternal deletion plus neuronal antisense silencing of the paternal allele; an AAV transgene under a neuronal promoter restores UBE3A expression independently of both silenced endogenous alleles

Evidence source: Mabb AM et al. 2011 Trends Neurosci PMID 21592595

Study-Level Limitations

- Catalog-relative ranking: current catalog contains 21 precedent programs and 8 vectors, so absence of a strong match is not proof that no therapy is possible.
 - Modality coverage is limited mainly to AAV and lentiviral precedents; dual-AAV, LNP/mRNA, genome editing, ASO, and transplant-enabling strategies are not fully represented.
 - Endpoint risk: CNS/neurodevelopmental outcomes may require natural-history data, age-stratified endpoints, and long follow-up because short-term clinical change can be hard to interpret.
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Top 5 GT Precedent Matches

Rank	Program	Vector	Score	Confidence	Approval
1	OAV101-IT	AAV9	7.1/10	Medium	approved
2	Zolgensma	AAV9	7.1/10	Medium	approved

Rank	Program	Vector	Score	Confidence	Approval
3	BMN 307	AAV5	6.9/10	Medium	phase2
4	Libmeldy	LV	6.9/10	Medium	approved
5	Skysona	LV	6.9/10	Medium	approved

Match #1: OAV101-IT

Precedent disease: Spinal Muscular Atrophy

Vector: AAV9

Tissue target: CNS/spinal cord

Composite score: 7.1 / 10

Score Breakdown

Dimension	Score	Max	What it measures
Packaging fit	1.50	2.0	Gene CDS size vs vector cargo capacity
Tissue tropism	2.00	2.0	Vector naturally reaches disease target tissue
Protein class	1.50	2.0	Same secreted/lysosomal/membrane/intracellular class
Pathway similarity	1.00	2.0	Same or related biological pathway
Modality compatibility	1.50	2.0	Disease mechanism supports gene-addition precedent
Inheritance compatibility	0.30	1.0	AR/XL loss-of-function pattern match
Approval precedent	1.00	1.0	Regulatory approval / trial stage
Immunogenicity	1.00	2.0	Pre-existing NAb seroprevalence for this vector
Therapeutic window	1.50	2.0	Can GT be given before irreversible damage?
Cross-correction	0.20	1.0	Can transduced cells rescue untransduced neighbours?
Immune privilege	0.90	1.0	Immunological protection of target tissue
Promoter availability	0.80	1.0	Validated tissue-specific promoters exist
Route of administration	0.70	1.0	Established delivery route to target tissue
Organelle targeting	1.00	1.0	Nuclear AAV delivery reaches correct subcellular compartment
TOTAL (normalised)	7.10	10.0	Raw sum / 21 × 10 (v2: 14 dimensions)

Rationale

- Gene CDS 2598bp / cargo 4700bp (55% utilized)
- Vector tropism plus precedent target match: cns

- Both intracellular proteins
- Inheritance mismatch (dominant/mitochondrial — higher complexity)
- Unknown pathway — neutral score
- Disease mechanism: genomic imprinting — Maternal UBE3A is lost by deletion or mutation; paternal UBE3A is epigenetically silenced in neurons by the SNHG14/UBE3A-ATS antisense transcript; neuronal UBE3A expression is absent from both alleles
- Gene-addition modality compatibility: conditionally supports gene addition; review disease-specific constraints
- Mechanism evidence: Mabb AM et al. (2011 Trends Neurosci PMID 21592595) described the dual-allele loss mechanism — maternal deletion plus neuronal antisense silencing of the paternal allele; an AAV transgene under a neuronal promoter restores UBE3A expression independently of both silenced endogenous alleles
- Mechanism source: Mabb AM et al. 2011 Trends Neurosci PMID 21592595 (<https://pubmed.ncbi.nlm.nih.gov/21592595/>)
- Approval status: approved
- Vector immunogenicity (AAV9): high (~22%) — substantial patient exclusion expected; immunodepletion protocols may be needed
- Moderate therapeutic window — progressive disease with childhood onset; early intervention strongly recommended; newborn screening integration beneficial
- Intracellular or membrane-bound protein — no cross-correction possible; each target cell must individually receive the vector; requires high transduction efficiency and therefore a higher or more targeted dose
- Immune privilege: high privilege — blood-brain barrier severely limits T-cell access; durable expression expected
- Promoter availability: Synapsin-1 (pan-neuronal), CaMKII (excitatory neurons), GFAP (astrocytes) — validated but cell-type specificity varies
- Route of administration: Intrathecal or ICV delivery — more invasive than IV; established in OAV101-IT and RGX-121 but higher procedural risk
- ORGANELLE TARGETING: COMPATIBLE — UBE3A standard subcellular localisation; nuclear AAV delivery directly produces functional protein at the correct cellular compartment with no additional organelle-import steps required.

Manual Review Flags

- Inheritance/mechanism may not be simple loss-of-function replacement; check dominant-negative, gain-of-function, or mitochondrial biology
- Mechanism evidence is conditionally compatible with gene addition; review the listed disease-specific constraints before treating vector precedent as transferable
- Dominant inheritance flagged; assess whether silencing, editing, or allele-specific strategy is needed instead of simple addition
- AAV tropism is species- and route-dependent; confirm human target-cell biodistribution rather than relying on animal tropism alone

Match #2: Zolgensma

Precedent disease: Spinal Muscular Atrophy

Vector: AAV9

Tissue target: CNS/motor neuron

Composite score: 7.1 / 10

Score Breakdown

Dimension	Score	Max	What it measures
Packaging fit	1.50	2.0	Gene CDS size vs vector cargo capacity
Tissue tropism	2.00	2.0	Vector naturally reaches disease target tissue
Protein class	1.50	2.0	Same secreted/lysosomal/membrane/intracellular class
Pathway similarity	1.00	2.0	Same or related biological pathway
Modality compatibility	1.50	2.0	Disease mechanism supports gene-addition precedent
Inheritance compatibility	0.30	1.0	AR/XL loss-of-function pattern match
Approval precedent	1.00	1.0	Regulatory approval / trial stage
Immunogenicity	1.00	2.0	Pre-existing NAb seroprevalence for this vector
Therapeutic window	1.50	2.0	Can GT be given before irreversible damage?
Cross-correction	0.20	1.0	Can transduced cells rescue untransduced neighbours?
Immune privilege	0.90	1.0	Immunological protection of target tissue
Promoter availability	0.80	1.0	Validated tissue-specific promoters exist
Route of administration	0.70	1.0	Established delivery route to target tissue
Organelle targeting	1.00	1.0	Nuclear AAV delivery reaches correct subcellular compartment
TOTAL (normalised)	7.10	10.0	Raw sum / 21×10 (v2: 14 dimensions)

Rationale

- Gene CDS 2598bp / cargo 4700bp (55% utilized)
- Vector tropism plus precedent target match: cns
- Both intracellular proteins
- Inheritance mismatch (dominant/mitochondrial — higher complexity)
- Unknown pathway — neutral score
- Disease mechanism: genomic imprinting — Maternal UBE3A is lost by deletion or mutation; paternal UBE3A is epigenetically silenced in neurons by the SNHG14/UBE3A-ATS antisense transcript; neuronal UBE3A expression is absent from both alleles
- Gene-addition modality compatibility: conditionally supports gene addition; review disease-specific constraints
- Mechanism evidence: Mabb AM et al. (2011 Trends Neurosci PMID 21592595) described the dual-allele loss mechanism — maternal deletion plus neuronal antisense silencing of the paternal allele; an AAV transgene under a neuronal promoter restores UBE3A expression independently of both silenced endogenous alleles
- Mechanism source: Mabb AM et al. 2011 Trends Neurosci PMID 21592595 (<https://pubmed.ncbi.nlm.nih.gov/21592595/>)
- Approval status: approved
- Vector immunogenicity (AAV9): high (~22%) — substantial patient exclusion expected; immunodepletion protocols may be needed

- Moderate therapeutic window — progressive disease with childhood onset; early intervention strongly recommended; newborn screening integration beneficial
- Intracellular or membrane-bound protein — no cross-correction possible; each target cell must individually receive the vector; requires high transduction efficiency and therefore a higher or more targeted dose
- Immune privilege: high privilege — blood-brain barrier severely limits T-cell access; durable expression expected
- Promoter availability: Synapsin-1 (pan-neuronal), CaMKII (excitatory neurons), GFAP (astrocytes) — validated but cell-type specificity varies
- Route of administration: Intrathecal or ICV delivery — more invasive than IV; established in OAV101-IT and RGX-121 but higher procedural risk
- ORGANELLE TARGETING: COMPATIBLE — UBE3A standard subcellular localisation; nuclear AAV delivery directly produces functional protein at the correct cellular compartment with no additional organelle-import steps required.

Manual Review Flags

- Inheritance/mechanism may not be simple loss-of-function replacement; check dominant-negative, gain-of-function, or mitochondrial biology
- Mechanism evidence is conditionally compatible with gene addition; review the listed disease-specific constraints before treating vector precedent as transferable
- Dominant inheritance flagged; assess whether silencing, editing, or allele-specific strategy is needed instead of simple addition
- AAV tropism is species- and route-dependent; confirm human target-cell biodistribution rather than relying on animal tropism alone

Match #3: BMN 307

Precedent disease: Phenylketonuria

Vector: AAV5

Tissue target: liver

Composite score: 6.9 / 10

Score Breakdown

Dimension	Score	Max	What it measures
Packaging fit	1.50	2.0	Gene CDS size vs vector cargo capacity
Tissue tropism	1.00	2.0	Vector naturally reaches disease target tissue
Protein class	1.50	2.0	Same secreted/lysosomal/membrane/intracellular class
Pathway similarity	1.00	2.0	Same or related biological pathway
Modality compatibility	1.50	2.0	Disease mechanism supports gene-addition precedent
Inheritance compatibility	0.30	1.0	AR/XL loss-of-function pattern match
Approval precedent	0.60	1.0	Regulatory approval / trial stage
Immunogenicity	2.00	2.0	Pre-existing NAb seroprevalence for this vector

Dimension	Score	Max	What it measures
Therapeutic window	1.50	2.0	Can GT be given before irreversible damage?
Cross-correction	0.20	1.0	Can transduced cells rescue untransduced neighbours?
Immune privilege	0.90	1.0	Immunological protection of target tissue
Promoter availability	0.80	1.0	Validated tissue-specific promoters exist
Route of administration	0.70	1.0	Established delivery route to target tissue
Organelle targeting	1.00	1.0	Nuclear AAV delivery reaches correct subcellular compartment
TOTAL (normalised)	6.90	10.0	Raw sum / 21×10 (v2: 14 dimensions)

Rationale

- Gene CDS 2598bp / cargo 4700bp (55% utilized)
- Vector tropism overlaps cns, but precedent target is liver
- Both intracellular proteins
- Inheritance mismatch (dominant/mitochondrial — higher complexity)
- Unknown pathway — neutral score
- Disease mechanism: genomic imprinting — Maternal UBE3A is lost by deletion or mutation; paternal UBE3A is epigenetically silenced in neurons by the SNHG14/UBE3A-ATS antisense transcript; neuronal UBE3A expression is absent from both alleles
- Gene-addition modality compatibility: conditionally supports gene addition; review disease-specific constraints
- Mechanism evidence: Mabb AM et al. (2011 Trends Neurosci PMID 21592595) described the dual-allele loss mechanism — maternal deletion plus neuronal antisense silencing of the paternal allele; an AAV transgene under a neuronal promoter restores UBE3A expression independently of both silenced endogenous alleles
- Mechanism source: Mabb AM et al. 2011 Trends Neurosci PMID 21592595 (<https://pubmed.ncbi.nlm.nih.gov/21592595/>)
- Approval status: phase2
- Vector immunogenicity (AAV5): low (~9%) — most patients eligible; minimal screening burden
- Moderate therapeutic window — progressive disease with childhood onset; early intervention strongly recommended; newborn screening integration beneficial
- Intracellular or membrane-bound protein — no cross-correction possible; each target cell must individually receive the vector; requires high transduction efficiency and therefore a higher or more targeted dose
- Immune privilege: high privilege — blood-brain barrier severely limits T-cell access; durable expression expected
- Promoter availability: Synapsin-1 (pan-neuronal), CaMKII (excitatory neurons), GFAP (astrocytes) — validated but cell-type specificity varies
- Route of administration: Intrathecal or ICV delivery — more invasive than IV; established in OAV101-IT and RGX-121 but higher procedural risk
- ORGANELLE TARGETING: COMPATIBLE — UBE3A standard subcellular localisation; nuclear AAV delivery directly produces functional protein at the correct cellular compartment with no additional organelle-import steps required.

Manual Review Flags

- Only partial tissue match; verify target-cell transduction and delivery route manually
- Inheritance/mechanism may not be simple loss-of-function replacement; check dominant-negative, gain-of-function, or mitochondrial biology
- Mechanism evidence is conditionally compatible with gene addition; review the listed disease-specific constraints before treating vector precedent as transferable
- Dominant inheritance flagged; assess whether silencing, editing, or allele-specific strategy is needed instead of simple addition
- AAV tropism is species- and route-dependent; confirm human target-cell biodistribution rather than relying on animal tropism alone

Match #4: Libmeldy

Precedent disease: Metachromatic leukodystrophy

Vector: LV

Tissue target: hematopoietic/CNS

Composite score: 6.9 / 10

Score Breakdown

Dimension	Score	Max	What it measures
Packaging fit	1.50	2.0	Gene CDS size vs vector cargo capacity
Tissue tropism	1.50	2.0	Vector naturally reaches disease target tissue
Protein class	0.50	2.0	Same secreted/lysosomal/membrane/intracellular class
Pathway similarity	1.00	2.0	Same or related biological pathway
Modality compatibility	1.50	2.0	Disease mechanism supports gene-addition precedent
Inheritance compatibility	0.30	1.0	AR/XL loss-of-function pattern match
Approval precedent	1.00	1.0	Regulatory approval / trial stage
Immunogenicity	2.00	2.0	Pre-existing NAb seroprevalence for this vector
Therapeutic window	1.50	2.0	Can GT be given before irreversible damage?
Cross-correction	0.20	1.0	Can transduced cells rescue untransduced neighbours?
Immune privilege	0.90	1.0	Immunological protection of target tissue
Promoter availability	0.80	1.0	Validated tissue-specific promoters exist
Route of administration	0.70	1.0	Established delivery route to target tissue
Organelle targeting	1.00	1.0	Nuclear AAV delivery reaches correct subcellular compartment
TOTAL (normalised)	6.86	10.0	Raw sum / 21×10 (v2: 14 dimensions)

Rationale

- Gene CDS 2598bp / cargo 8000bp (32% utilized)
- Precedent target match: cns
- Protein class mismatch
- Inheritance mismatch (dominant/mitochondrial — higher complexity)
- Unknown pathway — neutral score
- Disease mechanism: genomic imprinting — Maternal UBE3A is lost by deletion or mutation; paternal UBE3A is epigenetically silenced in neurons by the SNHG14/UBE3A-ATS antisense transcript; neuronal UBE3A expression is absent from both alleles
- Gene-addition modality compatibility: conditionally supports gene addition; review disease-specific constraints
- Mechanism evidence: Mabb AM et al. (2011 Trends Neurosci PMID 21592595) described the dual-allele loss mechanism — maternal deletion plus neuronal antisense silencing of the paternal allele; an AAV transgene under a neuronal promoter restores UBE3A expression independently of both silenced endogenous alleles
- Mechanism source: Mabb AM et al. 2011 Trends Neurosci PMID 21592595 (<https://pubmed.ncbi.nlm.nih.gov/21592595/>)
- Approval status: approved
- Vector immunogenicity (LV): low (~2%) — most patients eligible; minimal screening burden
- Moderate therapeutic window — progressive disease with childhood onset; early intervention strongly recommended; newborn screening integration beneficial
- Intracellular or membrane-bound protein — no cross-correction possible; each target cell must individually receive the vector; requires high transduction efficiency and therefore a higher or more targeted dose
- Immune privilege: high privilege — blood-brain barrier severely limits T-cell access; durable expression expected
- Promoter availability: Synapsin-1 (pan-neuronal), CaMKII (excitatory neurons), GFAP (astrocytes) — validated but cell-type specificity varies
- Route of administration: Intrathecal or ICV delivery — more invasive than IV; established in OAV101-IT and RGX-121 but higher procedural risk
- ORGANELLE TARGETING: COMPATIBLE — UBE3A standard subcellular localisation; nuclear AAV delivery directly produces functional protein at the correct cellular compartment with no additional organelle-import steps required.

Manual Review Flags

- Vector does not naturally cover all annotated disease tissues: cns
- Inheritance/mechanism may not be simple loss-of-function replacement; check dominant-negative, gain-of-function, or mitochondrial biology
- Mechanism evidence is conditionally compatible with gene addition; review the listed disease-specific constraints before treating vector precedent as transferable
- Dominant inheritance flagged; assess whether silencing, editing, or allele-specific strategy is needed instead of simple addition

Match #5: Skysona

Precedent disease: Cerebral adrenoleukodystrophy

Vector: LV

Tissue target: hematopoietic/CNS

Composite score: 6.9 / 10

Score Breakdown

Dimension	Score	Max	What it measures
Packaging fit	1.50	2.0	Gene CDS size vs vector cargo capacity
Tissue tropism	1.50	2.0	Vector naturally reaches disease target tissue
Protein class	0.50	2.0	Same secreted/lysosomal/membrane/intracellular class
Pathway similarity	1.00	2.0	Same or related biological pathway
Modality compatibility	1.50	2.0	Disease mechanism supports gene-addition precedent
Inheritance compatibility	0.30	1.0	AR/XL loss-of-function pattern match
Approval precedent	1.00	1.0	Regulatory approval / trial stage
Immunogenicity	2.00	2.0	Pre-existing NAb seroprevalence for this vector
Therapeutic window	1.50	2.0	Can GT be given before irreversible damage?
Cross-correction	0.20	1.0	Can transduced cells rescue untransduced neighbours?
Immune privilege	0.90	1.0	Immunological protection of target tissue
Promoter availability	0.80	1.0	Validated tissue-specific promoters exist
Route of administration	0.70	1.0	Established delivery route to target tissue
Organelle targeting	1.00	1.0	Nuclear AAV delivery reaches correct subcellular compartment
TOTAL (normalised)	6.86	10.0	Raw sum / 21×10 (v2: 14 dimensions)

Rationale

- Gene CDS 2598bp / cargo 8000bp (32% utilized)
- Precedent target match: cns
- Protein class mismatch
- Inheritance mismatch (dominant/mitochondrial — higher complexity)
- Unknown pathway — neutral score
- Disease mechanism: genomic imprinting — Maternal UBE3A is lost by deletion or mutation; paternal UBE3A is epigenetically silenced in neurons by the SNHG14/UBE3A-ATS antisense transcript; neuronal UBE3A expression is absent from both alleles
- Gene-addition modality compatibility: conditionally supports gene addition; review disease-specific constraints
- Mechanism evidence: Mabb AM et al. (2011 Trends Neurosci PMID 21592595) described the dual-allele loss mechanism — maternal deletion plus neuronal antisense silencing of the paternal allele; an AAV transgene under a neuronal promoter restores UBE3A expression independently of both silenced endogenous alleles
- Mechanism source: Mabb AM et al. 2011 Trends Neurosci PMID 21592595 (<https://pubmed.ncbi.nlm.nih.gov/21592595/>)
- Approval status: approved
- Vector immunogenicity (LV): low (~2%) — most patients eligible; minimal screening burden

- Moderate therapeutic window — progressive disease with childhood onset; early intervention strongly recommended; newborn screening integration beneficial
- Intracellular or membrane-bound protein — no cross-correction possible; each target cell must individually receive the vector; requires high transduction efficiency and therefore a higher or more targeted dose
- Immune privilege: high privilege — blood-brain barrier severely limits T-cell access; durable expression expected
- Promoter availability: Synapsin-1 (pan-neuronal), CaMKII (excitatory neurons), GFAP (astrocytes) — validated but cell-type specificity varies
- Route of administration: Intrathecal or ICV delivery — more invasive than IV; established in OAV101-IT and RGX-121 but higher procedural risk
- ORGANELLE TARGETING: COMPATIBLE — UBE3A standard subcellular localisation; nuclear AAV delivery directly produces functional protein at the correct cellular compartment with no additional organelle-import steps required.

Manual Review Flags

- Vector does not naturally cover all annotated disease tissues: cns
- Inheritance/mechanism may not be simple loss-of-function replacement; check dominant-negative, gain-of-function, or mitochondrial biology
- Mechanism evidence is conditionally compatible with gene addition; review the listed disease-specific constraints before treating vector precedent as transferable
- Dominant inheritance flagged; assess whether silencing, editing, or allele-specific strategy is needed instead of simple addition